

深圳市百年芯科技有限公司

Shen Zhen BaiNianXin Technology Co., Ltd.

SPECIFICATION

Product Type: 10.1"TFT LCD Module

LCD Nunmber: 1024×600

BNX Module No. : B101B545C-27A

CUSTOMER APPROVED	PREPARE BY	CHECK BY	APPROVED BY
SUPPLIER APPROVED	PREPARE BY	CHECK BY	APPROVED BY
	Kim		GuangEn Jin

1.0 General Description

1.1 Introduction

BNX Display model B101B545C-27A is a color active matrix thin film transistor (TFT) liquid crystal display without polarizer. This model is composed of amorphous silicon TFT as a switching device.

This TFT LCD has a 10.1-inch wide (16:9) diagonally measured active display area with WVGA (1024 horizontal by 600 vertical pixel) resolution. Each pixel is divided into Red, Green, Blue dots which are arranged in vertical stripes.

1.2 Features

- 10.1(16:9 diagonal)inch configuration
- 16.7M
- ROHS Compliance & Halogen Free

1.3 Applications

- Tablet
- Multimedia applications and Others AV system

1.4 General information

Item	Specification	Unit
Outline Dimension	235 (H) x143 (V) x3.5 (D)	mm
Display area	222.72 (H) ×125.28 (V)	mm
Number of Pixel	1024(H) × 3(RGB)×600(V)	pixels
Pixel pitch	0.0725(H)×0.2088 (V)	mm
Pixel arrangement	RGB Vertical stripe	
Display Mode	Normally Black	

2.0 ABSOLUTE MAXIMUM RATINGS

(The following are maximum values, which if exceeded, may cause operation or damage to the unit.)

Item	Symbol	Min.	Max.	Unit	Note
LC Operating Voltage	V _{OP}	--	4.5	V	*1,*2
Operating Temperature	T _{OP}	-20	70	°C	
Storage Temperature	T _{ST}	-30	80	°C	
Operating Ambient Humidity	H _{OP}	10	*4	RH	*3
Storage Humidity	H _{ST}	10	*4	RH	*3

Note:

- *1. At 25±5°C
- *2. Due to the characteristics of LC Material, the Liquid Crystal driving voltage varies with environmental temperature.
- *3. Non-condensation.
- *4. Temp.≤ 60°C ,90%RH Max.
Temp. > 60°C ,Absolute humidity shall be less than 90%RH.

3.0 Electrical Specifications

Item	Symbol	Min.	Typ.	Max.	Unit	Note
Analog Power	AVDD	9.0	9.6	10.2	V	
TFT Gate ON Voltage	VGH	17	18	19	V	*1
TFT Gate OFF Voltage	VGL	-5.5	-6	-6.5	V	*2
TFT Common Voltage	Vcom	3.9	4.2	4.5	V	*3
Data (RGB signal) Voltage	Vsig	0.1	--	4.5	V	
Power Supply Input Voltage	VDD	2.8	3.2	3.6	V	

Note:

*1. VGH is TFT Gate operating Voltage.

*2. VGL is TFT Gate operating Voltage.

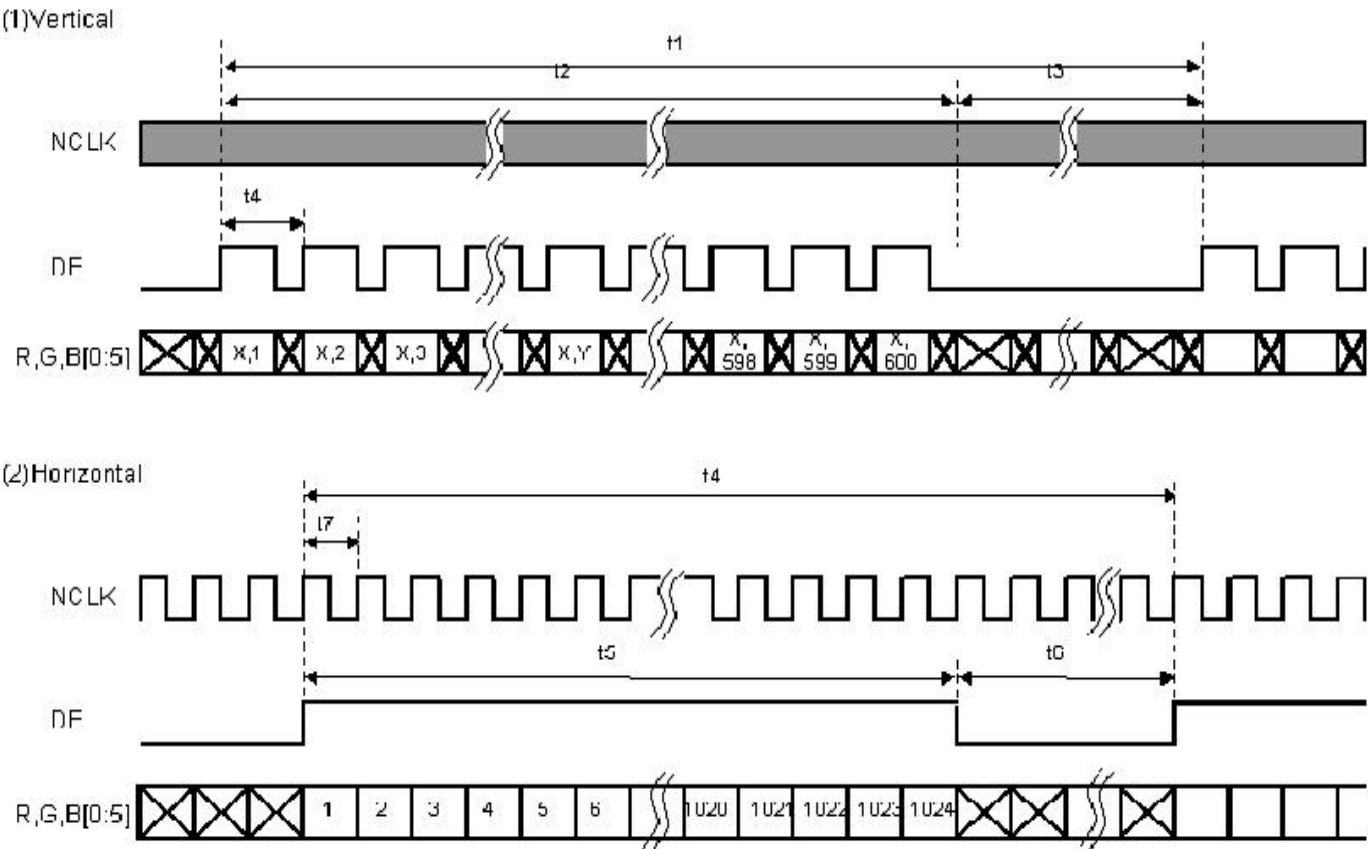
The storage structure of this model is C_{ST} (Storage on Common)

*3. Vcom must be adjusted to optimize display quality Crosstalk, Contrast Ratio and etc.

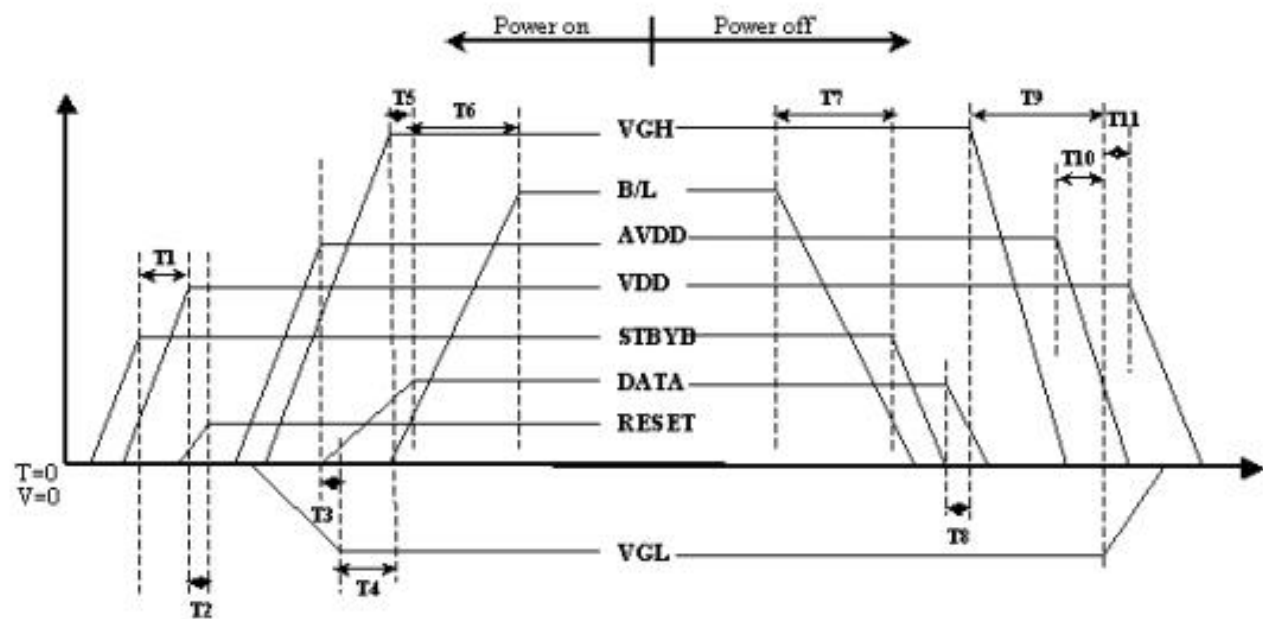
3.1 Interface Timing (DE mode)

Item	Symbol	Min.	Typ.	Max.	Unit
Frame Rate	--	55	60	65	Hz
Frame Period	t1	612	635	680	line
Vertical Display Time	t2	600	600	600	line
Vertical Blanking Time	t3	12	35	80	line
1 Line Scanning Time	t4	1114	1344	1400	clock
Horizontal Display Time	t5	1024	1024	1024	clock
Horizontal Blanking Time	t6	90	320	376	clock
Clock Rate	t7	40.8	51.2	61.6	MHz

Timing Diagram of Interface Signal (DE mode)



3.2 Power On/Off Sequence



Item	Min.	Typ.	Max.	Unit
T1	0	--	--	ms
T2	50	--	--	ms
T3	5	--	--	ms
T4	10	--	--	ms
T5	20	--	--	ms
T6	50	--	--	ms
T7	20	--	--	ms
T8	10	--	--	ms
T9	20	--	--	ms
T10	10	--	--	ms
T11	20	--	--	ms

Note (1) LED life time (Hr) can be defined as the time in which it continues to operate under the condition: $T_a=25\pm3\text{ }^{\circ}\text{C}$, typical IL value indicated in the above table until the brightness becomes less than 50%.

4.0 Optical Characteristics

Parameter		Symbol	Condition	Min.	Typ.	Max.	Unit	Remark
Viewing Angle range	Horizontal	Θ_3	CR > 10	-	85	-	Deg.	WV-Pol Note 1
		Θ_9		-	85	-	Deg.	
	Vertical	Θ_{12}		-	85	-	Deg.	
		Θ_6		-	85	-	Deg.	
Luminance Contrast ratio		CR	$\Theta = 0^\circ$	-	800	-		Note 2
Cell Transmittance		Tr		4.8	5.8	-	%	
White Chromaticity		x_w		TYP. - 0.03	0.307	TYP. + 0.03		Note 4 Base on C Light
		y_w			0.338			
Reproduction of color (C light)	Red	R_x			0.605			
		R_y			0.336			
	Green	G_x			0.297			
		G_y			0.552			
	Blue	B_x			0.139			
		B_y			0.132			
Color Gamut (C light)					-	50	-	%
Response Time (Rising + Falling)		T _{RT}	Ta= 25° C $\Theta = 0^\circ$	-	30	40	ms	Note 3
LCMluminance(Center)		YL	I=180mA		220	250	—	cd/m ²

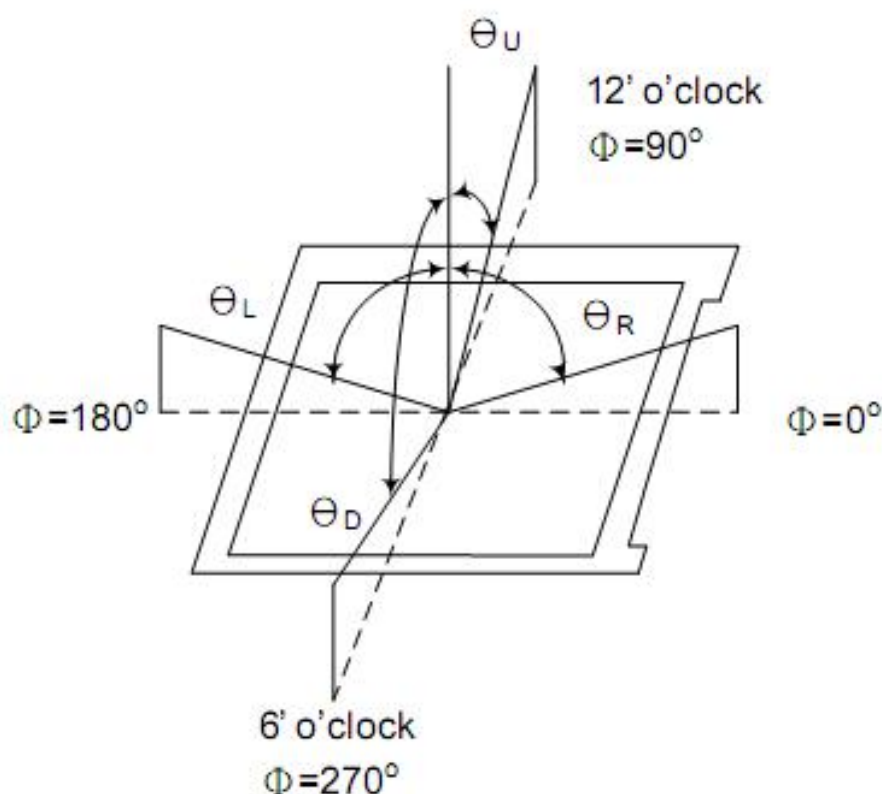
4.1 Measuring Condition

- Measuring surrounding : dark room .
- Ambient temperature : $25 \pm 2^\circ \text{C}$.
- 30min . Warm-up time .

4.2 Measuring Equipment

- FPM520 of Westar Display technologies, INC., which utilized SR-3 for Chromaticity and BM-5A for other optical characteristics (or related equipments).
- Measuring spot size : 20 ~ 21 mm

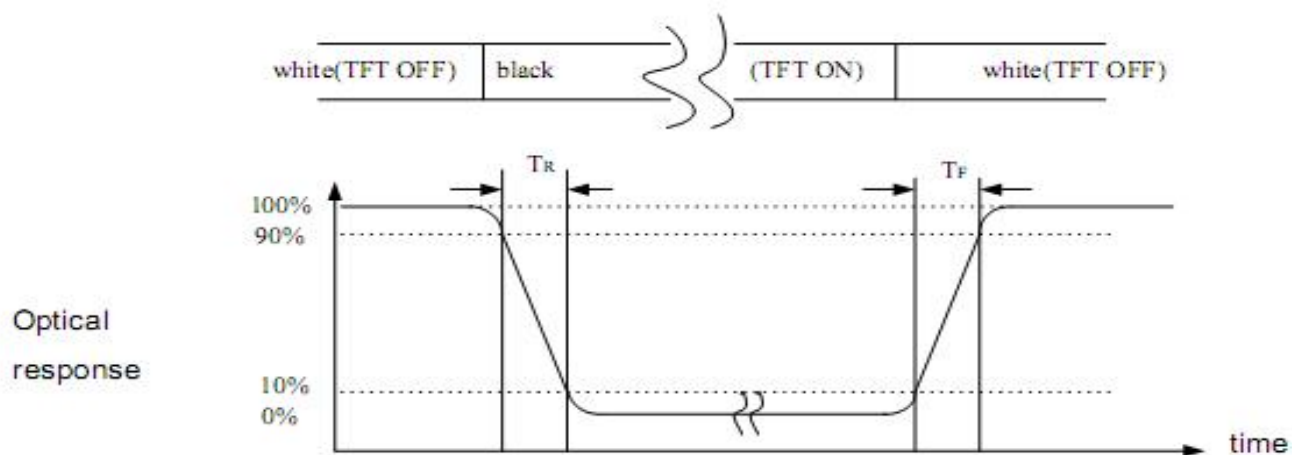
Note (1) Definition of Viewing Angle:



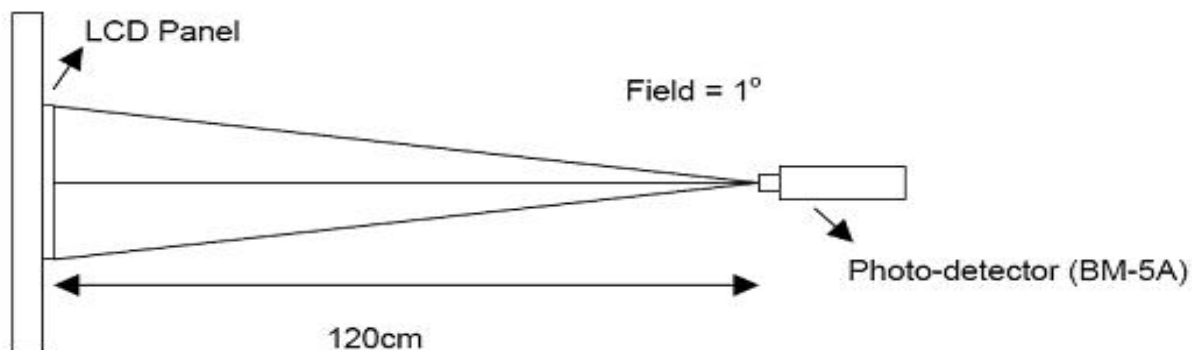
Note (2) Definition of Contrast Ratio (CR) :
measured at the center point of panel

$$CR = \frac{\text{Luminance with all pixels white}}{\text{Luminance with all pixels black}}$$

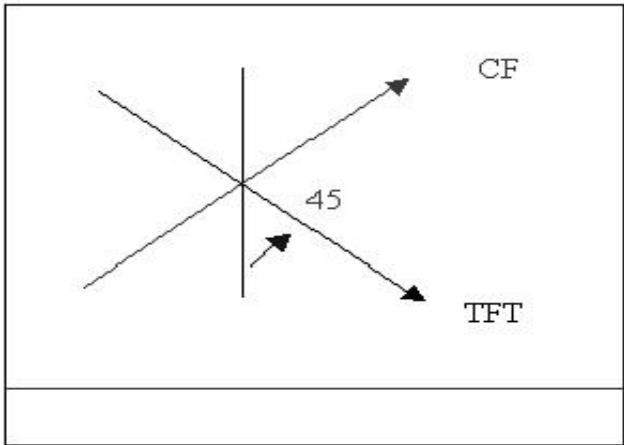
Note (3) Definition of Response Time : Sum of T_R and T_F



Note (4) Definition of optical measurement setup

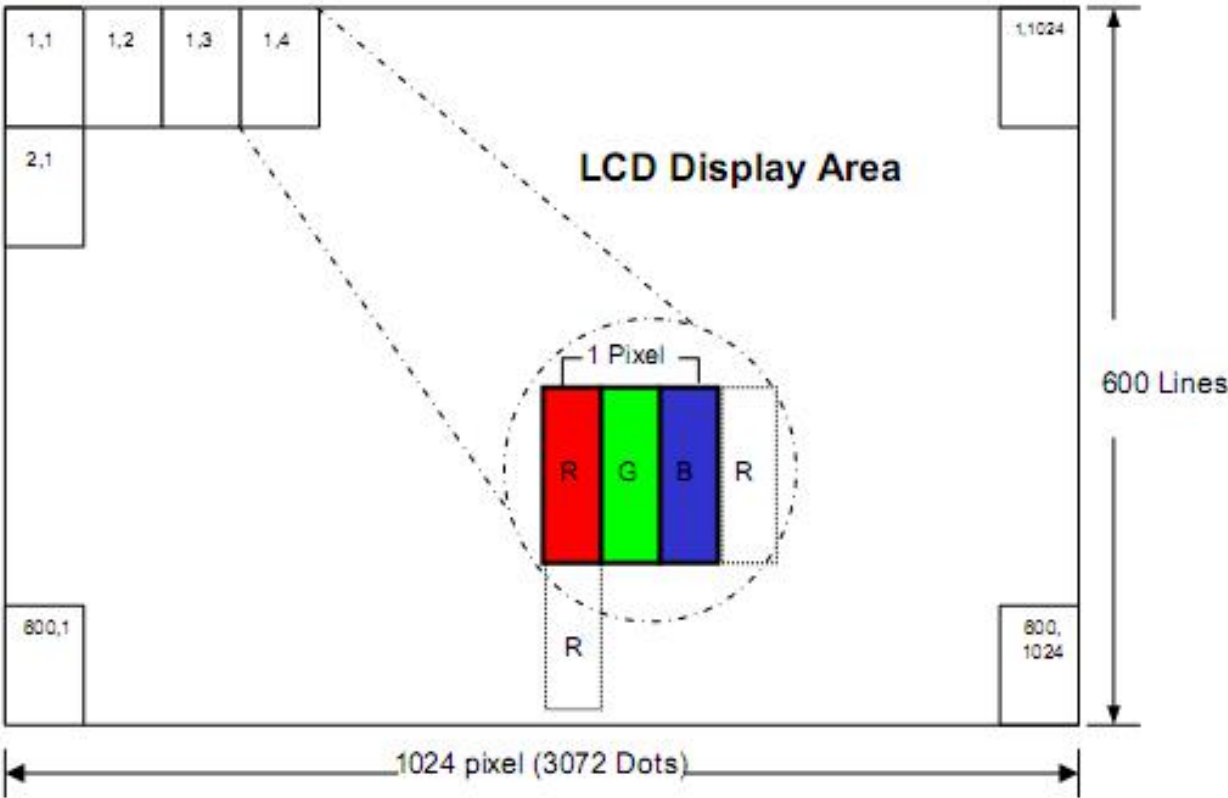


Note (5) Rubbing Direction (The different Rubbing Direction will cause the different optima view direction.



(Alignment Direction)

5.0 Pixel Format



5.3 Relationship Between Displayed Color and Input

5.3.1 6bit

	Display	MSB LSB R5 R4 R3 R2 R1 R0	M L G5 G4 G3 G2 G1 G0	S S G5 G4 G3 G2 G1 G0	B B B5 B4 B3 B2 B1 B0	Gray scale level
Basic color	Black	L L L L L L L	L L L L L L L	L L L L L L L	L L L L L L L	-
	Blue	L L L L L L L	L L L L L L L	H H H H H H H	L L L L L L L	-
	Green	L L L L L L L	H H H H H H H	L L L L L L L	L L L L L L L	-
	Light Blue	L L L L L L L	H H H H H H H	H H H H H H H	L L L L L L L	-
	Red	H H H H H H H	L L L L L L L	L L L L L L L	L L L L L L L	-
	Purple	H H H H H H H	L L L L L L L	L L L L L L L	H H H H H H H	-
	Yellow	H H H H H H H	H H H H H H H	L L L L L L L	L L L L L L L	-
	White	H H H H H H H	H H H H H H H	H H H H H H H	H H H H H H H	-
Gray scale of Red	Black	L L L L L L L	L L L L L L L	L L L L L L L	L L L L L L L	L0
	Dark ↑ ↓ Light	L L L L L L H	L L L L L L L	L L L L L L L	L L L L L L L	L1
		L L L L L H L	L L L L L L L	L L L L L L L	L L L L L L L	L2
		:	:	:	:	L3...L60
	Light	H H H H L H L	L L L L L L L	L L L L L L L	L L L L L L L	L61
		H H H H H L L	L L L L L L L	L L L L L L L	L L L L L L L	L62
		H H H H H H H	L L L L L L L	L L L L L L L	L L L L L L L	Red L63
	Red	H H H H H H H	L L L L L L L	L L L L L L L	L L L L L L L	Red L63
Gray scale of Green	Black	L L L L L L L	L L L L L L L	L L L L L L L	L L L L L L L	L0
	Dark ↑ ↓ Light	L L L L L L L	L L L L L L L	L L L L L L L	L L L L L L L	L1
		L L L L L L L	L L L L L L L	L L L L L L L	L L L L L L L	L2
		:	:	:	:	L3...L60
	Light	L L L L L L L	H H H H L H L	L L L L L L L	L L L L L L L	L61
		L L L L L L L	H H H H H L L	L L L L L L L	L L L L L L L	L62
		L L L L L L L	H H H H H H H	L L L L L L L	L L L L L L L	Green L63
	Green	L L L L L L L	H H H H H H H	L L L L L L L	L L L L L L L	Green L63
Gray scale of Blue	Black	L L L L L L L	L L L L L L L	L L L L L L L	L L L L L L L	L0
	Dark ↑ ↓ Light	L L L L L L L	L L L L L L L	L L L L L L L	L L L L L L L	L1
		L L L L L L L	L L L L L L L	L L L L L L L	L L L L L L L	L2
		:	:	:	:	L3...L60
	Light	L L L L L L L	L L L L L L L	L L L L L L L	H H H H L H L	L61
		L L L L L L L	L L L L L L L	L L L L L L L	H H H H H L L	L62
		L L L L L L L	L L L L L L L	L L L L L L L	H H H H H H H	Blue L63
	Blue	L L L L L L L	L L L L L L L	L L L L L L L	H H H H H H H	Blue L63
Gray scale of White & Black	Black	L L L L L L L	L L L L L L L	L L L L L L L	L L L L L L L	L0
	Dark ↑ ↓ Light	L L L L L L H	L L L L L L L	L L L L L L L	L L L L L L L	L1
		L L L L L H L	L L L L L L L	L L L L L L L	L L L L L L L	L2
		:	:	:	:	L3...L60
	Light	H H H H L H L	H H H H L H L	H H H H L H L	H H H H L H L	L61
		H H H H H L L	H H H H H L L	H H H H H L L	H H H H H L L	L62
		H H H H H H H	H H H H H H H	H H H H H H H	H H H H H H H	White L63
	White	H H H H H H H	H H H H H H H	H H H H H H H	H H H H H H H	White L63

5.3.1 8bit

[illegible]

6.0 Interface Connection:

Pin No	Symbol	I/O	Function	Remark
1	VLED+	P	Power for LED backlight (Anode)	
2	VLED+	P	Power for LED backlight (Anode)	
3	VLED-	P	Power for LED backlight (Cathode)	
4	VLED-	P	Power for LED backlight (Cathode)	
5	GND	P	Power ground	
6	Vcom	I	Common voitage	
7	DVdd	P	Ppwer for Digital Circuit	
8	MODE	I	DE/SYNC mode select	Note 1
9	DE	I	Data input Enable	
10	VS	I	Vertical Sync input	
11	HS	I	Horizontal Sync input	
12	B7	I	Blue data(MSB)	
13	B6	I	Blue data	
14	B5	I	Blue data	
15	B4	I	Blue data	
16	B3	I	Blue data	
17	B2	I	Blue data	
18	B1	I	Blue data	Note 2
19	B0	I	Blue data(LSB)	Note 2
20	G7	I	Green data(MSB)	
21	G6	I	Green data	
22	G5	I	Green data	
23	G4	I	Green data	
24	G3	I	Green data	
25	G2	I	Green data	
26	G1	I	Green data	Note 2

27	G0	I	Green data (LSB)	Note 2
28	R7	I	Red data (MSB)	
29	R6	I	Red data	
30	R5	I	Red data	
31	R4	I	Red data	
32	R3	I	Red data	
33	R2	I	Red data	
34	R1	I	Red data	Note 2
35	R0	I	Red data (LSB)	Note 2
36	GND	P	Power Ground	
37	DCLK	I	Sample clock	Note 3
38	GND	P	Power Ground	
39	L/R	I	Left/right selection pin	
40	U/D	I	Up/down selection	
41	Vgh	P	Gate ON Voltage	
42	Vgl	P	Gate OFF Voltage	
43	Avdd	P	Power for Analog Circuit	
44	RESET	I	Global Reset pin	
45	NC	—	No connection	
46	Vcom	I	Common Voltage	
47	DLTHB	I	Dithering Function	
48	GND	P	Power Ground	
49	NC	—	No connection	
50	NC	—	No connection	

1:input, 0:output, P:Power

Note 1:DE/SYNC mode select. Normally pull high.

When select DE mode, MODE="1", VS and HS must pull high.

When select SYNC mode, MODE="0", DE must be grounded.

Note 2:When input 18 bits RGB data, the two low bits of R, G and B and must be grounded.

Note 3:Data shall be latched at the falling edge of DCLK.

5. Outline dimension



	MIN	TYP	MAX
x	0.28	0.30	0.32
y	0.30	0.32	0.34

12. RoHs must be complied. (Use Lead-free process)



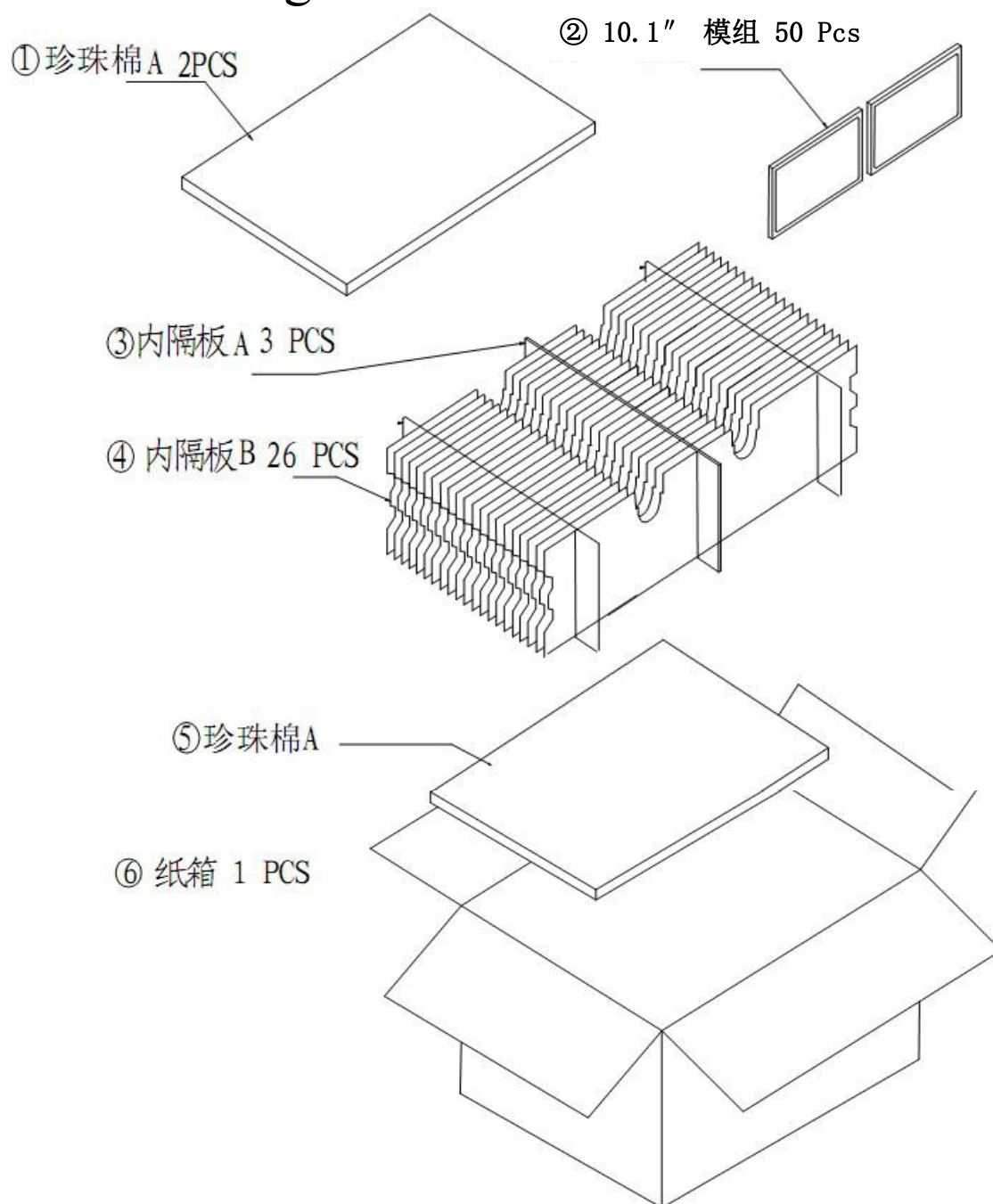
8.0 RELIABILITY TEST

No	Test Items	Conditions	Note
1	High Temperature Storage	Ta=+80℃ 120hrs	
2	Low Temperature Storage	Ta=-30℃ 120hrs	
3	High Temperature Operation	Ta=+70℃ 120hrs	
4	Low Temperature Operation	Ta=-20℃ 120hrs	
5	High Temperature High Humidity Operation	Ta=+60℃ ;90%RH 120hrs	

Note :(1)All tests above are practiced at module type

(2)There is no display function NG issue occurred , all the cosmetic specification is judged before the reliability stress.

9.0 Packing form



10.0 GENERAL PRECAUTION

10.1 Use Restriction

This product is not authorized for use in life supporting systems, aircraft navigation control systems, military systems and any other application where performance failure could be life-threatening or otherwise catastrophic.

10.2 Disassembling or Modification

Do not disassemble or modify the LCD. It may damage sensitive parts inside LCD, and may cause scratches or dust on the display. HannStar does not warrant the LCD, if customers disassemble or modify the module.

10.3 Breakage of LCD Panel

10.3.1 If LCD panel is broken and liquid crystal spills out, do not ingest or inhale liquid crystal, and do not contact liquid crystal with skin.

10.3.2 If liquid crystal contacts mouth or eyes, rinse out with water immediately.

10.3.4 If liquid crystal contacts skin or cloths, wash it off immediately with alcohol and rinse thoroughly with water.

10.3.4 Handle carefully with chips of glass that may cause injury, when the glass is broken.

10.4 Absolute Maximum Ratings and Power Protection Circuit

10.4.1 Do not exceed the absolute maximum rating values, such as the supply voltage variation, input voltage variation, variation in parts' parameters, environmental temperature, etc., otherwise LCD may be damaged.

10.4.2 Please do not leave LCD in the environment of high humidity and high temperature for a long time.

10.4.3 It's recommended to employ protection circuit for power supply.

10.5 Operation

10.5.1 Do not touch, push or rub the polarizer with anything harder than HB pencil lead if the LCD attaches a polarizer.

10.5.2 Use fingerstalls or soft gloves in order to keep clean display quality, when persons handle the LCD for incoming inspection or assembly.

10.5.3 When the surface is dusty, please wipe gently with absorbent cotton or other soft material.

10.5.4 Wipe off saliva or water drops as soon as possible. If saliva or water drops contact with polarizer for a long time, they may cause deformation or color fading.

10.5.5 When cleaning the adhesives, please use absorbent cotton wetted with a little petroleum benzine or other adequate solvent.

10.6 Static Electricity

- 10.8.1 Protection film must remove very slowly from the surface of LCD to prevent from electrostatic occurrence if the LCD attaches a polarizer.
- 10.8.2 Because TFT-LCD panel is very weak to electrostatic discharge, please be careful with electrostatic discharge.
Persons who handle the LCD should be grounded through adequate methods.

10.7 Strong Light Exposure

The LCD shall not be exposed under strong light such as direct sunlight. Otherwise display characteristics may be changed.

10.8 Disposal

When disposing LCD, obey the local environmental regulations.